BLOOD RELATIONS

Family Trees, Direct Relations and Coded Relations

OAVS TGT | Reasoning (Additional topic from the “etc.” in the syllabus) | Topic 19

Syllabus line (Reasoning, 30 Marks): The verbal and non-verbal topics listed in the syllabus end with “etc.”. This topic is not named in that line, but it is asked in the papers: coded blood relations appear in Mathematics 2024 (Q94) and Science 2024 (Q92).

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). Two blood-relation questions appear (Maths 2024 Q94, Science 2024 Q92). Both are solved below. The practice questions were generated or checked by program so that each has exactly one correct option.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	The 2 PYQs, the coding used, priority list
2	Foundation	Relation table, gender clues, coded-relation method
3	Solved PYQs	2 PYQs with step-by-step reasoning
4	Shortcuts, Traps and Scope	Speed methods and mistakes
5	Practice Set	6 original questions with a verified key
6	Quick Revision	One-page cheat sheet

CHAPTER 1

PYQ Analysis

Two coded-relation questions

Paper	Q.	Question	Symbols used	Ans
Mathematics 2024	94	Which means U is the mother’s brother’s son of Z?	+ son, − brother, × wife, ÷ father	A
Science 2024	92	How is P related to K if P + R ÷ G @ H # K?	+ son, # sister, ÷ wife, @ father	4 (Brother)
English 2019 / English 2022 / Social Studies 2023	–	none		

PYQ INSIGHT: Both PYQs are **coded relations**: four symbols stand for four relations, and a chain of letters must be read from left to right as a family tree. Maths 2024 asks you to **find the correct code** for a given relation; Science 2024 asks you to **find the relation** from a given code. Expect one coded-relation question in every paper.

Pattern	PYQs	Priority
Coded relations: read a chain and find the relation	1	★★★ Must
Coded relations: choose the code for a given relation	1	★★★ Must
Direct relations (“pointing to a girl, he said …”)	0	★★ Expected
Family tree with several people	0	★★ Expected

CHAPTER 2

Foundation

Relation table, gender clues and the chain method

2.1 Relations you must know

Term	Meaning	Gender
Paternal uncle	Father’s brother	Male
Paternal aunt	Father’s sister	Female
Maternal uncle	Mother’s brother	Male
Maternal aunt	Mother’s sister	Female
Cousin	Child of an uncle or aunt	Not fixed
Nephew / niece	Son / daughter of a brother or sister	Male / female
Brother-in-law	Spouse’s brother, sister’s husband, or spouse’s sister’s husband	Male
Sister-in-law	Spouse’s sister, brother’s wife	Female
Son-in-law / daughter-in-law	Daughter’s husband / son’s wife	Male / female
Grandfather / grandmother	Parent’s father / mother	Male / female

NOTE: Words that do **not** fix gender: cousin, sibling, parent, child, spouse, grandchild, friend. If the gender is not given, the relation cannot be fixed (“data inadequate”).

2.2 Coded relations: the method

Step	Do this
1	Write down what each symbol means (for example + son, − brother, × wife, ÷ father).
2	Translate the chain pair by pair, from left to right: “A + B” means A is the son of B.
3	Note the gender each pair gives (son: male, wife: female, father: male, sister: female).
4	Draw the tree: one line per generation, spouses side by side, siblings on the same line.
5	Read the relation between the first and the last person, or compare the options one by one.

2.3 Short forms that save time

Expression	Equals
My father’s only son	Myself (or my brother if I am a girl)
My mother’s only daughter	Myself (if I am a girl) or my sister
Father’s brother’s son	Cousin
Mother’s brother	Maternal uncle
Mother’s mother’s only daughter	Mother
Sister’s husband	Brother-in-law

SHORTCUT: Work **backwards** from the answer: for “choose the code for U is the mother’s brother’s son of Z”, write the relation as a chain (Z’s mother, her brother, his son) and match it to the options.

CHAPTER 3

Solved PYQs

Both questions with step-by-step reasoning

Mathematics 2024 Q94

In a certain code language, ‘A + B’ means A is the son of B, ‘A − B’ means A is the brother of B, ‘A × B’ means A is the wife of B, ‘A ÷ B’ means A is the father of B. Which of the following means that U is the mother’s brother’s son of Z? (A) U + V − W × Y ÷ Z (B) U + V − W × Y − Z (C) U + V − W × Y + Z (D) U + V − W + Y ÷ Z

1. Write the relation as a chain: Z’s mother, then her brother, then his son (U).
2. Test option A: U + V means U is the son of V; V − W means V is the brother of W; W × Y means W is the wife of Y; Y ÷ Z means Y is the father of Z.
3. So W is Z’s mother (wife of Z’s father Y), V is W’s brother, so V is Z’s mother’s brother, and U is V’s son. That is exactly the relation asked.
4. Option B makes Y a brother of Z; option C makes Y a son of Z; option D makes W a son of Y, so W could not be a wife. Only A fits.
Answer: Option A

Science 2024 Q92

A + B means A is the son of B; A # B means A is the sister of B; A ÷ B means A is the wife of B; A @ B means A is the father of B. How is P related to K if ‘P + R ÷ G @ H # K’? (1) Father (2) Husband (3) Son (4) Brother

1. Translate pair by pair: P + R (P is the son of R); R ÷ G (R is the wife of G); G @ H (G is the father of H); H # K (H is the sister of K).
2. H and K are brother and sister, so G is also the father of K, and R (G’s wife) is the mother of K.
3. P is the son of R, so P is a son of the same parents as K. P is male (son), so P is K’s **brother**.
4. Father and husband are ruled out because P is the son of R and R is the wife of G; son is ruled out because P and K are children of the same parents.
Answer: Option 4 (Brother)

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and mistakes

• **Translate in order:** never skip a link; each “A symbol B” gives the relation of A to B only.

• **Gender from the symbol:** son, brother, husband, father are male; wife, sister, mother, daughter are female.

• **Same parent test:** if two people are both the child of one person, they are siblings (cousins need different parents).

• **Use the options:** in “choose the code” questions, test the option that starts with the given person.

• **Time:** 45 seconds per question.

Trap	How to avoid it
Reading the chain backwards	A + B means A is the son of B, so the first person is the younger.
Assuming gender	Cousin, sibling, child do not fix gender.
Mixing up sides	Father’s side (paternal) and mother’s side (maternal) are different relations.
Calling a spouse’s sibling a sibling	Spouse’s brother is a brother-in-law.
Ignoring “only”	“My father’s only son” is myself (or my brother), not a different person.

Scope

Related PYQs / topics	Where
Ranking, seating and direction questions	Their own earlier topics
Data sufficiency of a family puzzle with many people	Blood Relations (family trees) here

CHAPTER 5

Practice Set

6 original questions; answers on the next page

A. Direct relations

Q1. Pointing to a man, a woman said, “His mother is the only daughter of my mother.” How is the woman related to the man?

A. Sister B. Aunt C. Mother D. Grandmother

Q2. A is the brother of B. B is the sister of C. C is the father of D. How is A related to D?

A. Father B. Uncle C. Cousin D. Brother

Q3. Pointing to a girl, Sameer said, “She is the daughter of my father’s only son.” How is the girl related to Sameer?

A. Daughter B. Sister C. Niece D. Cousin

Q4. Anil’s mother’s brother’s wife has a daughter, Rina. How is Rina related to Anil?

A. Sister B. Aunt C. Niece D. Cousin

B. Coded relations

Q5. A % B means A is the daughter of B; A $ B means A is the father of B; A # B means A is the wife of B; A @ B means A is the brother of B. How is Q related to M in ‘M % N # P @ Q’?

A. Father B. Maternal uncle C. Paternal uncle D. Cousin

Q6. A % B means A is the daughter of B; A $ B means A is the father of B; A # B means A is the wife of B; A @ B means A is the brother of B. How is N related to K in ‘K % L $ M # N’?

A. Brother-in-law B. Brother C. Uncle D. Cousin

Answer key

Q	Answer	Reason
1	C	The only daughter of the woman’s mother is the woman herself. So the man’s mother is the woman: she is his mother.
2	B	A, B and C are brothers and a sister. C is D’s father, so A is the brother of D’s father: a paternal uncle.
3	A	Sameer’s father’s only son is Sameer himself, so the girl is his daughter.
4	D	Anil’s mother’s brother is his maternal uncle. The uncle’s wife’s daughter is the uncle’s daughter, so Rina is Anil’s cousin.
5	C	M % N: M is the daughter of N. N # P: N is the wife of P, so P is M’s father. P @ Q: P is the brother of Q. So Q is the brother of M’s father: a paternal uncle.
6	A	K % L: K is the daughter of L. L $ M: L is the father of M. M # N: M is the wife of N. K and M are sisters (both daughters of L), so N, the husband of M, is K’s brother-in-law.

CHAPTER 6

Quick Revision

One page for the last day

Item	Rule
Coded chain	Translate each pair in order; first person is the younger
Gender	Son, brother, father, husband: male. Daughter, sister, mother, wife: female
Same parent	Children of one person are siblings
Only	My father’s only son = me (or my brother)
Brother-in-law	Sister’s husband; spouse’s brother

PYQ RULE: Both PYQs are coded relations: translate each pair, note gender, then find the relation between the first and last person (Science 2024) or the option that gives the required chain (Maths 2024).

ROUTINE: Exam-day routine: decode the symbols; translate in order; fix gender; draw the tree; match the answer. 45 seconds per question.